

A STABILIZER-FREE WEAK VIRTUAL ELEMENT METHOD FOR THE STOKES PROBLEM ON POLYTOPAL MESHES*

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1. Introduction. A discretization of incompressible Stokes flow must preserve mass and control the velocity through its symmetric gradient. On polygonal and polyhedral meshes, these requirements call for discrete spaces with suitable normal and tangential traces. The virtual element method provides such spaces on general polytopes. Its degrees of freedom determine the polynomial projections needed to assemble the discrete problem without evaluating the virtual basis functions [1]. In standard formulations, a stabilization term controls the part of a virtual function that its polynomial projection does not capture.

The $H(\text{div})$ -conforming virtual elements of Beirão da Veiga, Brezzi, Marini, and Russo [2] provide continuous normal traces and polynomial divergence. Chen and Wang [6] combined these elements with tangential face unknowns and a weak symmetric gradient to obtain a divergence-free weak virtual element method for the Stokes problem. Their formulation includes a stabilization term that controls the boundary mismatch between the projected cell velocity and its traces. This framework gives a natural starting point for a method whose weak strain alone controls that mismatch.

Stabilizer-free methods achieve stability through the choice of the local reconstruction space. Mu, Ye, and Zhang [8] developed a stabilizer-free weak Galerkin method for the Stokes equations on polytopal meshes. For second-order elliptic problems, Chen, Huang, and Wei [7] constructed virtual element methods without extrinsic stabilization by using stable gradient projections onto macro finite element spaces. These results motivate local reconstructions that retain enough information to control the discrete energy. For the symmetric-gradient formulation of Stokes flow, the reconstruction must also account for rigid motions and the coupling of normal and tangential velocity traces.

Here we develop a stabilizer-free weak virtual element method in two and three dimensions. The velocity consists of an $H(\text{div})$ virtual field and a tangential polynomial field on the primal mesh faces. The pressure is a discontinuous polynomial of degree $k - 1$. We reconstruct the weak strain in a symmetric tensor space that is polynomial on the cones joining each element's star center to its faces. The strain is computed locally from the velocity degrees of freedom. The tensor space provides independent face moments and volume moments that control both the symmetric gradient of the projected velocity and the boundary mismatch. The discrete viscous energy is the squared L^2 norm of this weak strain.

Under the stated mesh regularity and a uniform spanning condition on the face-tangential symmetric tensors, we prove norm equivalence, coercivity, and the velocity–pressure inf-sup condition. The discrete velocity is exactly divergence-free on every primal element. For sufficiently smooth solutions, we obtain order- h^k error bounds in

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the discrete velocity norm and the L^2 pressure norm. The spanning condition is an explicit assumption of the analysis. The numerical section reports two-dimensional tests and identifies the rectangular-mesh results as tests outside this assumption.

Section 2 introduces the meshes, velocity spaces, and projections. Section 3 constructs the weak strain and proves its stability. Section 4 presents the discrete problem and error estimates. Section 5 gives the numerical results.

2. VEM spaces. Let $\Omega \subset \mathbb{R}^d$, $d = 2, 3$ be a bounded and simply connected polytopal domain. Given an external force field $\mathbf{f} \in \mathbf{L}^2(\Omega)$, we consider the steady-state incompressible Stokes problem:

$$(2.1) \quad \begin{cases} -\operatorname{div}(\varepsilon(\mathbf{u})) + \nabla p = \mathbf{f} & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \\ \mathbf{u} = \mathbf{0} & \text{on } \partial\Omega. \end{cases}$$

Here $\varepsilon(\mathbf{u}) := (\nabla \mathbf{u} + \nabla \mathbf{u}^\top)/2$ is the symmetric strain, and the pressure is normalized by $\int_\Omega p \, dx = 0$. We use $(\cdot, \cdot)_K$ and $\langle \cdot, \cdot \rangle_{\partial K}$ for the volume and boundary L^2 inner products, and $\|\cdot\|_K$ and $\|\cdot\|_{\partial K}$ for their norms.

2.1. Primal and dual meshes. Let $\mathcal{K}_h$ be a conforming polytopal mesh of Ω . Each element K is star-shaped with respect to a ball of radius at least ρh_K , where $h_K = \operatorname{diam} K$ and $\rho > 0$ is independent of $h := \max_K h_K$. Let $\mathbf{x}_K$ be the center of such a ball. We assume a uniformly bounded number of faces per element. In three dimensions, each face is also uniformly star-shaped. The cones $T = \{\mathbf{x}_K + t(\mathbf{y} - \mathbf{x}_K) : \mathbf{y} \in F, 0 \leq t \leq 1\}$ form a refinement $\mathcal{T}_h$; they are triangles in two dimensions and pyramids in three dimensions. We assume that these cones admit a conforming, uniformly shape-regular simplicial refinement with simplex diameters comparable to h_K . In particular, $h_F := \operatorname{diam} F$ is comparable to h_K . These assumptions are used in the trace, inverse, and scaling estimates below.

Let $\mathcal{F}_h$ and $\mathcal{F}_h^T$ be the face sets of $\mathcal{K}_h$ and $\mathcal{T}_h$, respectively. Their interior faces are denoted by $\mathring{\mathcal{F}}_h^K$ and $\mathring{\mathcal{F}}_h^T$. Let $\mathcal{T}_h(K)$ be the set of cones in K .

Let $F \in \partial K$ be a face with unit normal $\mathbf{n}_F$ and orthonormal tangent vectors $\{\mathbf{t}_i^F\}_{i=1}^{d-1}$. For any $\mathbf{v} \in \mathbb{R}^d$, the tangential part on F is

$$\Pi_F \mathbf{v} = (\mathbf{I} - \mathbf{n}_F \otimes \mathbf{n}_F) \mathbf{v} = \sum_{i=1}^{d-1} (\mathbf{v} \cdot \mathbf{t}_i^F) \mathbf{t}_i^F.$$

With $\operatorname{sym} A = (A + A^\top)/2$ and $\operatorname{skw} A = (A - A^\top)/2$, define

$$\mathcal{T}^F := \operatorname{span}\{\mathbf{t}_1^F, \dots, \mathbf{t}_{d-1}^F\},$$

$$\mathcal{T}^F(\mathbb{S}) := \operatorname{span}\{\operatorname{sym}(\mathbf{t}_i^F \otimes \mathbf{t}_j^F) \mid 1 \leq i \leq j \leq d-1\},$$

$$\mathcal{N}^F(\mathbb{S}) := \operatorname{sym}(\mathcal{T}^F \otimes \mathbf{n}_F) \oplus \operatorname{span}\{\mathbf{n}_F \otimes \mathbf{n}_F\},$$

$$\mathcal{T}^F(\mathbb{K}) := \operatorname{span}\{\operatorname{skw}(\mathbf{t}_i^F \otimes \mathbf{t}_j^F) \mid 1 \leq i < j \leq d-1\}.$$

These subspaces give the following $\mathbf{t}$ - $\mathbf{n}$ decompositions

$$(2.2) \quad \mathbb{S} = \mathcal{T}^F(\mathbb{S}) \oplus \mathcal{N}^F(\mathbb{S}),$$

$$(2.3) \quad \mathbb{K} = \mathcal{T}^F(\mathbb{K}) \oplus \mathcal{N}^F(\mathbb{K}), \quad \mathcal{N}^F(\mathbb{K}) = \operatorname{skw}(\mathcal{T}^F \otimes \mathbf{n}_F).$$

Fix an integer $k \geq 1$. For $K \in \mathcal{K}_h$, let $\mathbb{P}_k(K)$ be the space of polynomials on K of degree at most k . Let $\mathbb{P}_k(K; \mathbb{X}) = \mathbb{P}_k(K) \otimes \mathbb{X}$, where $\mathbb{X}$ is $\mathbb{R}^d$ for vectors, $\mathbb{S}$ for symmetric matrices, and $\mathbb{K}$ for skew-symmetric matrices.

We set $\mathbb{P}_j = \{0\}$ for $j < 0$, and $\mathbf{RM}(K) = \{\mathbf{a} + \mathbf{B}\mathbf{x} : \mathbf{a} \in \mathbb{R}^d, \mathbf{B} \in \mathbb{K}\}$.

2.2. VEM spaces. The local velocity space and its degrees of freedom are the $H(\text{div})$ virtual elements of [2]. On each element $K \in \mathcal{K}_h$, we introduce a local space

$$\mathbf{V}_K = \left\{ \mathbf{v} \in \mathbf{H}(\text{div}, K) \cap \mathbf{H}(\text{curl}, K) : \mathbf{v} \cdot \mathbf{n} \in \mathbb{P}_k(F) \ \forall F \subset \partial K, \right. \\ \left. \text{div } \mathbf{v} \in \mathbb{P}_{k-1}(K), \text{ curl } \mathbf{v} \in \text{curl } \mathbb{P}_k(K; \mathbb{R}^d) \right\},$$

The global virtual element space is

$$\mathbf{V}_h^{\text{div}} := \{ \mathbf{v} \in \mathbf{H}(\text{div}, \Omega), \mathbf{v}|_K \in \mathbf{V}_K \ \forall K \in \mathcal{K}_h, \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \partial\Omega \}.$$

The degrees of freedom (DoFs) for $\mathbf{V}_K$ are

$$\text{Type I} \quad \int_F \mathbf{v} \cdot \mathbf{n} q_k \, ds \quad \forall q_k \in \mathbb{P}_k(F), F \subset \partial K,$$

$$\text{Type II} \quad \int_K \mathbf{v} \cdot \mathbf{q}_{k-2} \, d\mathbf{x} \quad \forall \mathbf{q}_{k-2} \in \mathbf{G}_{k-2}(K),$$

$$\text{Type III} \quad \int_K \mathbf{v} \cdot \mathbf{q}_k \, d\mathbf{x} \quad \forall \mathbf{q}_k \in \mathbf{G}_k(K)^\perp,$$

where $\mathbf{G}_k(K) = \nabla \mathbb{P}_{k+1}(K)$. Here $\mathbf{G}_k(K)^\perp$ is its $L^2(K)$ -orthogonal complement in $\mathbb{P}_k(K; \mathbb{R}^d)$. Moments are taken against a basis of each listed space. These DoFs are unisolvent [2].

On each face $F \in \mathcal{F}_h$, we introduce a space

$$\mathbf{V}_F := \mathbb{P}_{k-1}(F; \mathcal{T}^F) + \mathbf{RM}(F),$$

where $\mathbf{RM}(F)$ denotes the space of rigid motions, i.e., $\mathbf{RM}(F) := \Pi_F \mathbf{RM}(K) = \mathbb{P}_0(F; \mathcal{T}^F) \oplus \mathbb{P}_0(F; \mathcal{T}^F(\mathbb{K}))\mathbf{x}$. The DoFs for functions in $\mathbf{V}_F$ are

$$\int_F \mathbf{v} \cdot \mathbf{q} \, d\mathbf{s} \quad \forall \mathbf{q} \in \mathbb{P}_{k-1}(F; \mathcal{T}^F) + \mathbf{RM}(F).$$

The tangential face space is

$$\mathbf{V}_h^t := \{ \mathbf{v}^t, \mathbf{v}^t|_F \in \mathbf{V}_F, F \in \mathcal{F}_h, \mathbf{v}^t = \mathbf{0} \text{ on } \partial\Omega \}.$$

The weak virtual element space is

$$\mathbf{V}_h := \left\{ \{ \mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t \}, \mathbf{v}_h^{\text{div}} \in \mathbf{V}_h^{\text{div}}, \mathbf{v}_h^t \in \mathbf{V}_h^t \right\}.$$

2.3. Projection operator. For $\mathbf{v} \in \mathbf{L}^2(K)$, let $\Pi_K^o \mathbf{v} \in \mathbb{P}_k(K; \mathbb{R}^d)$ satisfy

$$\int_K \Pi_K^o \mathbf{v} \cdot \mathbf{q} \, d\mathbf{x} = \int_K \mathbf{v} \cdot \mathbf{q} \, d\mathbf{x} \quad \forall \mathbf{q} \in \mathbb{P}_k(K; \mathbb{R}^d).$$

For $\mathbf{v} \in \mathbf{V}_K$, Type III DoFs give the moments in $\mathbf{G}_k(K)^\perp$. The remaining moments follow by integration by parts: for $p_{k+1} \in \mathbb{P}_{k+1}(K)$,

$$\int_K \mathbf{v} \cdot \nabla p_{k+1} \, d\mathbf{x} = - \int_K \text{div } \mathbf{v} p_{k+1} \, d\mathbf{x} + \int_{\partial K} \mathbf{v} \cdot \mathbf{n} p_{k+1} \, d\mathbf{s}.$$

The normal trace is determined by Type I DoFs. The divergence is determined by Type I and II DoFs through

$$(\text{div } \mathbf{v}, q)_K = -(\mathbf{v}, \nabla q)_K + \langle \mathbf{v} \cdot \mathbf{n}, q \rangle_{\partial K}, \quad q \in \mathbb{P}_{k-1}(K).$$

Thus every term above is computable [2]. We write $(\Pi_h^o \mathbf{v})|_K = \Pi_K^o(\mathbf{v}|_K)$.

3. Weak strain.

3.1. Weak strain space. Let $T \in \mathcal{T}_h(K)$ have base face F . The local weak strain space is

$$\begin{aligned} \Sigma_k(T) = & \mathbb{P}_k(T) \mathbf{n}_F \otimes \mathbf{n}_F \\ & \oplus \text{sym}(\mathbf{n}_F \otimes [\mathbb{P}_{k-1}(T; \mathcal{T}^F) + \mathbf{RM}(F)]) \\ & \oplus \mathbb{P}_{k-1}(T; \mathcal{T}^F(\mathbb{S})). \end{aligned} \quad (3.1)$$

Face polynomials, including $\mathbf{RM}(F)$, are extended constantly along lines normal to F . For $k = 1$, the middle block is $\text{sym}(\mathbf{n}_F \otimes \mathbf{RM}(F))$; for $k > 1$, it is $\text{sym}(\mathbf{n}_F \otimes \mathbb{P}_{k-1}(T; \mathcal{T}^F))$. Set

$$V_k(F) := \mathbb{P}_k(F) \mathbf{n}_F \oplus \mathbf{V}_F.$$

The normal trace $\text{tr}_F \boldsymbol{\sigma} := \boldsymbol{\sigma} \mathbf{n}_F|_F$ satisfies $\text{tr}_F \Sigma_k(T) = V_k(F)$. In particular, $\mathbb{P}_{k-1}(K; \mathbb{S})$ is contained in the piecewise strain space on K .

We define the local symmetric tensor space on K as the discontinuous polynomial space

$$(3.2) \quad \Sigma_h^{-1}(K) = \bigoplus_{i=1}^n \chi_{T_i} \Sigma_k(T_i) = \{ \boldsymbol{\tau} \in L^2(K; \mathbb{S}) : \boldsymbol{\tau}|_T \in \Sigma_k(T) \text{ for } T \in \mathcal{T}_h(K) \},$$

where χ_{T_i} is the characteristic function of T_i . We define the bubble space

$$\begin{aligned} \mathbb{B}_k(\text{div}, K; \mathbb{S}) = & \{ \boldsymbol{\tau} \in \Sigma_h^{-1}(K) : \boldsymbol{\tau} \mathbf{n}|_{\partial K} = \mathbf{0} \} \\ = & \bigoplus_{i=1}^n \lambda_{\mathbf{x}_K} \chi_{T_i} (\mathbb{P}_{k-1}(T_i) \mathbf{n}_i \otimes \mathbf{n}_i) \\ & \oplus \bigoplus_{i=1}^n \lambda_{\mathbf{x}_K} \chi_{T_i} \text{sym}(\mathbb{P}_{k-2}(T_i; \mathcal{T}^{F_i}) \otimes \mathbf{n}_i) \\ & \oplus \bigoplus_{i=1}^n \chi_{T_i} \mathbb{P}_{k-1}(T_i; \mathcal{T}^{F_i}(\mathbb{S})). \end{aligned} \quad (3.3)$$

Here F_i is the base face of T_i , $\mathbf{n}_i$ is its outward normal, and $\lambda_{\mathbf{x}_K}$ is the affine function on T_i that equals one at $\mathbf{x}_K$ and zero on F_i . The bubble space imposes no continuity across the internal faces of K .

Assumption 3.1. There exist $d(d+1)/2$ symmetric tensors b_ℓ such that

$$(3.4) \quad \mathbb{S} = \text{span} \{ b_\ell, \ell = 1, \dots, d(d+1)/2 \},$$

where $\mathcal{T}^{F_i}(\mathbb{S}) = \text{span} \{ b_i^m, m = 1, \dots, d(d-1)/2 \}$ for $F_i \in \partial K$, and

$$(3.5) \quad \{ b_\ell \} \subset \{ b_i^m, i = 1, \dots, n, m = 1, \dots, d(d-1)/2 \}.$$

On every element, the tensors b_ℓ can be chosen with $|b_\ell| = 1$ and

$$(3.6) \quad \sum_{\ell=1}^{d(d+1)/2} c_\ell^2 \lesssim \left| \sum_{\ell=1}^{d(d+1)/2} c_\ell b_\ell \right|^2 \quad \text{for all real coefficients } c_\ell,$$

with a constant independent of h and K . Here $|\cdot|$ is the Frobenius norm.

All subsequent stability and error estimates assume Assumption 3.1. Constants may depend on k , Ω , the mesh regularity, and this tensor-basis bound, but not on h .

For each selected b_ℓ , let T_ℓ be a cone whose base face has $b_\ell \in \mathcal{T}^{F_i}(\mathbb{S})$, and define

$$\mathcal{E}_K := \bigoplus_{\ell=1}^{d(d+1)/2} \chi_{T_\ell} \mathbb{P}_{k-1}(T_\ell) b_\ell \subset \mathbb{B}_k(\text{div}, K; \mathbb{S}).$$

In the following lemma, $\mathbb{B}_k \ominus \mathcal{E}_K$ denotes the $L^2(K)$ -orthogonal complement of $\mathcal{E}_K$ within the bubble space.

LEMMA 3.2. For $K \in \mathcal{K}_h$, the space $\Sigma_h^{-1}(K)$ is uniquely determined by the following DoFs:

$$(3.6a) \quad \int_F \boldsymbol{\sigma} \mathbf{n} \cdot \mathbf{q} \, ds, \quad \mathbf{q} \in V_k(F), F \in \partial K,$$

$$(3.6b) \quad \int_K \boldsymbol{\sigma} : \boldsymbol{\tau} \, dx, \quad \boldsymbol{\tau} \in \mathbb{B}_k(\text{div}, K; \mathbb{S}) \ominus \mathcal{E}_K,$$

$$(3.6c) \quad \int_K \boldsymbol{\sigma} : \boldsymbol{\tau} \, dx, \quad \boldsymbol{\tau} \in \mathbb{P}_{k-1}(K; \mathbb{S}),$$

where moments are taken against bases of the listed spaces.

Proof. Suppose all moments vanish. Since the normal trace belongs to $V_k(F)$, the face moments imply $\boldsymbol{\sigma} \mathbf{n} = 0$ on ∂K . Thus $\boldsymbol{\sigma}$ is a bubble. The second group of moments then gives $\boldsymbol{\sigma} \in \mathcal{E}_K$, so $\boldsymbol{\sigma} = \sum_\ell \chi_{T_\ell} p_\ell b_\ell$ with $p_\ell \in \mathbb{P}_{k-1}(T_\ell)$. Let b_ℓ^* be the dual tensor basis, satisfying $b_j : b_\ell^* = \delta_{j\ell}$. Extend p_ℓ as a polynomial to K and choose $\boldsymbol{\tau} = p_\ell b_\ell^*$ in (3.6c). Then $\|p_\ell\|_{T_\ell}^2 = 0$ for every ℓ , so $\boldsymbol{\sigma} = 0$. The trace map is onto $\bigoplus_{F \subset \partial K} V_k(F)$ and has the bubble space as its kernel. Also, $\dim \mathcal{E}_K = \dim \mathbb{P}_{k-1}(K; \mathbb{S})$. The number of moments therefore equals $\dim \Sigma_h^{-1}(K)$. This proves unisolvence. $\square$

Let

$$\Sigma_h^{-1} = \{\boldsymbol{\sigma} \in L^2(\Omega; \mathbb{S}) : \boldsymbol{\sigma}|_T \in \Sigma_k(T), T \in \mathcal{T}_h\}.$$

3.2. Weak strain. For any $\mathbf{v} = \{\mathbf{v}^{\text{div}}, \mathbf{v}^t\} \in \mathbf{V}_h$, we define $\varepsilon_h^w(\mathbf{v}) \in \Sigma_h^{-1}$ satisfying $\varepsilon_h^w(\mathbf{v})|_K = \varepsilon_K^w(\mathbf{v})$ and

$$(3.7) \quad \begin{aligned} (\varepsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K &= (\varepsilon(\Pi_K^o \mathbf{v}^{\text{div}}), \boldsymbol{\sigma})_K \\ &\quad + \langle (\mathbf{v}^{\text{div}} - \Pi_K^o \mathbf{v}^{\text{div}}) \cdot \mathbf{n}, (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{n} \rangle_{\partial K} \\ &\quad + \langle \mathbf{v}^t - \Pi_F \Pi_K^o \mathbf{v}^{\text{div}}, \Pi_F(\boldsymbol{\sigma} \mathbf{n}) \rangle_{\partial K} \end{aligned}$$

for all $\boldsymbol{\sigma} \in \Sigma_h^{-1}(K)$, with Π_F applied facewise. Since $\boldsymbol{\sigma}$ is discontinuous inside K , integration by parts must include the internal jumps of $\boldsymbol{\sigma} \mathbf{n}$. Dropping these terms does not give an equivalent definition.

The normal and tangential boundary jumps are

$$\begin{aligned} \mathcal{J}_{\partial K}^n(\mathbf{v}) &:= ((\mathbf{v}^{\text{div}} - \Pi_K^o \mathbf{v}^{\text{div}}) \cdot \mathbf{n}) \mathbf{n}, \\ \mathcal{J}_{\partial K}^t(\mathbf{v}) &:= \mathbf{v}^t - \Pi_F \Pi_K^o \mathbf{v}^{\text{div}}. \end{aligned}$$

Their sum is $\mathcal{J}_{\partial K}(\mathbf{v}) := \mathcal{J}_{\partial K}^n(\mathbf{v}) + \mathcal{J}_{\partial K}^t(\mathbf{v})$.

On the face $F \subset \partial K$, we define $(\Pi_{\partial K}^t \mathbf{v})|_F \in \mathbf{V}_F$ as

$$\int_F \Pi_{\partial K}^t \mathbf{v} \cdot \mathbf{q} \, ds = \int_F \mathbf{v}|_K \cdot \mathbf{q} \, ds \quad \forall \mathbf{q} \in \mathbf{V}_F.$$

Define the notation $\Pi_{\partial K}$ as

$$\Pi_{\partial K} \mathbf{v} := (\mathbf{v} \cdot \mathbf{n}) \mathbf{n} + \Pi_{\partial K}^t \mathbf{v}.$$

Since $\boldsymbol{\sigma} \mathbf{n}|_F \in V_k(F)$, (3.7) becomes

$$(\varepsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K = (\varepsilon(\Pi_K^o \mathbf{v}^{\text{div}}), \boldsymbol{\sigma})_K + \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}), \boldsymbol{\sigma} \mathbf{n} \rangle_{\partial K}.$$

For $\mathbf{v}_h \in \mathbf{V}_h$, define the mesh-dependent seminorm

$$\|\mathbf{v}_h\|_{1,h}^2 := \sum_{K \in \mathcal{K}_h} \|\mathbf{v}_h\|_{1,h,K}^2,$$

where the seminorm on each element is defined as

$$\|\mathbf{v}_h\|_{1,h,K}^2 := \|\varepsilon(\mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}})\|_K^2 + h_K^{-1} \|\mathbf{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K}^2.$$

LEMMA 3.3. *The mesh-dependent quantity $\|\mathbf{v}_h\|_{1,h}$ defines a norm on the virtual element space $\mathbf{V}_h$.*

Proof. If $\|\mathbf{v}_h\|_{1,h} = 0$, then on each K , $\mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}} = \mathbf{r}_K \in \mathbf{RM}(K)$, and

$$\mathbf{v}_h^{\text{div}} \cdot \mathbf{n} = \mathbf{r}_K \cdot \mathbf{n}, \quad \mathbf{v}_h^t = \Pi_F \mathbf{r}_K \quad \text{on each } F \subset \partial K.$$

The second identity uses $\Pi_F \mathbf{r}_K \in \mathbf{RM}(F)$. The normal component of $\mathbf{v}_h^{\text{div}}$ and the single-valued tangential field $\mathbf{v}_h^t$ show that neighboring rigid motions agree on their common face. A rigid motion that vanishes on a face is zero, so connectedness gives one global rigid motion $\mathbf{r}$. The boundary conditions give both $\mathbf{r} \cdot \mathbf{n} = 0$ and $\Pi_F \mathbf{r} = 0$ on boundary faces. Hence $\mathbf{r} = 0$. The normal face moments and all polynomial volume moments of $\mathbf{v}_h^{\text{div}}$ now vanish. Unisolvence of $\mathbf{V}_K$ gives $\mathbf{v}_h^{\text{div}} = 0$, and the tangential identity gives $\mathbf{v}_h^t = 0$. $\square$

We also need control of the full jump. For $\mathbf{w} = \mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}}$, Korn's inequality modulo rigid motions gives a $\mathbf{r}_K \in \mathbf{RM}(K)$ such that

$$h_K^{-1} \|\mathbf{w} - \mathbf{r}_K\|_K + \|\nabla(\mathbf{w} - \mathbf{r}_K)\|_K \lesssim \|\varepsilon(\mathbf{w})\|_K.$$

Since $\mathbf{\Pi}_{\partial K}^t$ reproduces $\Pi_F \mathbf{r}_K$, the trace inequality yields

$$h_K^{-1/2} \|\Pi_F \mathbf{w} - \mathbf{\Pi}_{\partial K}^t \mathbf{w}\|_{\partial K} \lesssim \|\varepsilon(\mathbf{w})\|_K.$$

The normal part of the jump is unchanged by $\mathbf{\Pi}_{\partial K}$. The unprojected tangential part has the same norm as the difference above. Thus

$$(3.8) \quad h_K^{-1/2} \|\mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K} \lesssim \|\mathbf{v}_h\|_{1,h,K}.$$

The jump of $\mathbf{w}$ across a primal face is the difference of its two full boundary jumps; on a boundary face, $\mathbf{w}$ is minus its full boundary jump. Applying the broken Korn inequality [5] on the simplicial refinement therefore gives the global equivalence

$$(3.9) \quad \|\mathbf{v}_h\|_{1,h}^2 \simeq \sum_{K \in \mathcal{K}_h} (\|\nabla \mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}}\|_K^2 + h_K^{-1} \|\mathbf{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K}^2).$$

This equivalence is global; it does not hold on an individual element.

LEMMA 3.4. *We have the discrete inf-sup condition*

$$(3.10) \quad \|\mathbf{v}\|_{1,h} \lesssim \sup_{0 \neq \boldsymbol{\sigma}_h \in \Sigma_h^{-1}} \frac{\sum_{K \in \mathcal{K}_h} (\varepsilon_K^w(\mathbf{v}), \boldsymbol{\sigma}_h)_K}{\|\boldsymbol{\sigma}_h\|_{0,h}} \quad \forall \mathbf{v} \in \mathbf{V}_h.$$

202 *Proof.* Here $\|\boldsymbol{\sigma}\|_{0,h}^2 := \sum_K \|\boldsymbol{\sigma}\|_K^2$. For fixed $\mathbf{v} \in \mathbf{V}_h$, use Lemma 3.2 to choose
 203 $\boldsymbol{\sigma}|_K \in \Sigma_h^{-1}(K)$ with

$$\begin{aligned} 204 \quad (\boldsymbol{\sigma}, \boldsymbol{\tau})_K &= (\varepsilon(\boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}), \boldsymbol{\tau})_K & \forall \boldsymbol{\tau} \in \mathbb{P}_{k-1}(K; \mathbb{S}), \\ \langle \boldsymbol{\sigma} \mathbf{n}, \mathbf{q} \rangle_F &= \langle h_K^{-1} \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}), \mathbf{q} \rangle_F & \forall \mathbf{q} \in V_k(F), F \subset \partial K, \end{aligned}$$

205 and with the moments (3.6b) set to zero. Face moments are prescribed separately on
 206 each element side, using its outward normal. Testing with the polynomial strain and
 207 projected jump gives

$$208 \quad \sum_K (\varepsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K = \|\mathbf{v}\|_{1,h}^2.$$

209 On the rescaled element, unisolvence, mesh regularity, and the uniform basis bound
 210 in Assumption 3.1 give

$$\begin{aligned} 211 \quad \|\boldsymbol{\sigma}\|_K &\lesssim \|\varepsilon(\boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}})\|_K + h_K^{-1/2} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v})\|_{\partial K}, \\ \|\boldsymbol{\sigma}\|_{0,h} &\lesssim \|\mathbf{v}\|_{1,h}. \end{aligned}$$

212 Substitute this tensor into the supremum in (3.10). The result follows for $\mathbf{v} \neq 0$; the
 213 zero case is immediate. $\square$

214 **LEMMA 3.5.** *For every $\mathbf{v}_h \in \mathbf{V}_h$,*

$$215 \quad \|\mathbf{v}_h\|_{1,h} \lesssim \|\varepsilon_h^w(\mathbf{v}_h)\|.$$

216 *Proof.* Apply Lemma 3.4 and the Cauchy–Schwarz inequality. $\square$

217 **4. Numerical scheme.** Let

$$218 \quad S_h := \{q_h \in L_0^2(\Omega) : q_h|_K \in \mathbb{P}_{k-1}(K) \text{ for all } K \in \mathcal{K}_h\}.$$

219 The discrete formulation of (2.1) seeks $\mathbf{u}_h = \{\mathbf{u}_h^{\text{div}}, \mathbf{u}_h^t\} \in \mathbf{V}_h$ and $p_h \in S_h$ such that

$$220 \quad (4.1) \quad \begin{cases} a_h(\mathbf{u}_h, \mathbf{v}_h) + b(\mathbf{v}_h, p_h) = (\mathbf{f}, \boldsymbol{\Pi}_h^o \mathbf{v}_h^{\text{div}}) & \forall \mathbf{v}_h \in \mathbf{V}_h, \\ b(\mathbf{u}_h, q_h) = 0 & \forall q_h \in S_h, \end{cases}$$

221 where

$$222 \quad a_h(\mathbf{u}_h, \mathbf{v}_h) := (\varepsilon_h^w(\mathbf{u}_h), \varepsilon_h^w(\mathbf{v}_h)), \quad b(\mathbf{v}_h, p_h) := -(\text{div } \mathbf{v}_h^{\text{div}}, p_h).$$

223 **LEMMA 4.1.** *For every $\mathbf{v}_h = \{\mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t\} \in \mathbf{V}_h$, there holds*

$$224 \quad \|\varepsilon_h^w(\mathbf{v}_h)\|^2 \lesssim \|\mathbf{v}_h\|_{1,h}^2.$$

225 *Proof.* Set $\boldsymbol{\sigma} = \varepsilon_K^w(\mathbf{v}_h)$ in (3.7). The Cauchy–Schwarz and polynomial trace
 226 inequalities give

$$\begin{aligned} 227 \quad \|\varepsilon_K^w(\mathbf{v}_h)\|_K^2 &\lesssim \left(\|\varepsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}})\|_K \right. \\ &\quad \left. + h_K^{-1/2} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K} \right) \|\varepsilon_K^w(\mathbf{v}_h)\|_K. \end{aligned}$$

228 Divide by the strain norm when it is nonzero, square, and sum over K . $\square$

LEMMA 4.2. For $\mathbf{u}_h, \mathbf{v}_h \in \mathbf{V}_h$,

$$|a_h(\mathbf{u}_h, \mathbf{v}_h)| \lesssim \|\mathbf{u}_h\|_{1,h} \|\mathbf{v}_h\|_{1,h},$$

$$a_h(\mathbf{u}_h, \mathbf{u}_h) \gtrsim \|\mathbf{u}_h\|_{1,h}^2.$$

LEMMA 4.3. For $\mathbf{v}_h \in \mathbf{V}_h$ and $q_h \in S_h$,

$$|b(\mathbf{v}_h, q_h)| \lesssim \|\mathbf{v}_h\|_{1,h} \|q_h\|.$$

Proof. Write $\mathbf{w} = \boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}}$. For $q \in \mathbb{P}_{k-1}(K)$, the volume projection and integration by parts give

$$(\text{div } \mathbf{v}_h^{\text{div}}, q)_K = (\text{div } \mathbf{w}, q)_K + \langle \mathcal{J}_{\partial K}^n(\mathbf{v}_h) \cdot \mathbf{n}, q \rangle_{\partial K}.$$

Use $q = \text{div } \mathbf{v}_h^{\text{div}}$, the polynomial trace inequality, and $|\text{div } \mathbf{w}| \leq \sqrt{d} |\varepsilon(\mathbf{w})|$. This gives $\|\text{div } \mathbf{v}_h^{\text{div}}\|_K \lesssim \|\mathbf{v}_h\|_{1,h,K}$. Sum over K and apply the Cauchy–Schwarz inequality. $\square$

LEMMA 4.4. For every $p_h \in S_h$, there is a $\mathbf{v}_h \in \mathbf{V}_h$ with

$$\text{div } \mathbf{v}_h^{\text{div}} = p_h, \quad \|\mathbf{v}_h\|_{1,h} \lesssim \|p_h\|.$$

4.1. Interpolation. For $\mathbf{v} \in \mathbf{H}^1(K)$, define $\mathbf{I}_K^{\text{div}} \mathbf{v} \in \mathbf{V}_K$ by matching its Type I–III DoFs with those of $\mathbf{v}$. Integration by parts with $q \in \mathbb{P}_{k-1}(K)$ gives the commuting identity

$$(4.2) \quad \text{div } \mathbf{I}_K^{\text{div}} \mathbf{v} = Q_h(\text{div } \mathbf{v})|_K,$$

where Q_h is the elementwise L^2 projection onto $\mathbb{P}_{k-1}$; for mean-zero functions its range is S_h . The interpolant reproduces $\mathbb{P}_k(K; \mathbb{R}^d)$. Scaled stability of its DoFs gives

$$(4.3) \quad \|\mathbf{I}_K^{\text{div}} \mathbf{v}\|_K \lesssim \|\mathbf{v}\|_K + h_K |\mathbf{v}|_{1,K}.$$

Indeed, the scaled face moments are bounded by $h_K^{1/2} \|\mathbf{v} \cdot \mathbf{n}\|_{\partial K}$, and the normalized volume moments by $\|\mathbf{v}\|_K$. The trace inequality and uniform norm equivalence for the unisolvent DoFs prove (4.3). These arguments are the usual face-VEM interpolation estimates; see [2, 3] for the underlying spaces and related stability results. Subtracting a degree- k polynomial approximant in (4.3) yields

$$(4.4) \quad \|\mathbf{v} - \mathbf{I}_K^{\text{div}} \mathbf{v}\|_K \lesssim h_K^{k+1} |\mathbf{v}|_{k+1,K}.$$

Proof of Lemma 4.4. The continuous divergence right inverse [4] provides $\mathbf{v} \in \mathbf{H}_0^1(\Omega)$ with $\text{div } \mathbf{v} = p_h$ and $|\mathbf{v}|_1 \lesssim \|p_h\|$. Set $\mathbf{v}_h^{\text{div}}|_K = \mathbf{I}_K^{\text{div}} \mathbf{v}$ and $\mathbf{v}_h^t|_F = \boldsymbol{\Pi}_{\partial K}^t \mathbf{v}$. The traces are single-valued and zero on the boundary, and (4.2) gives the required divergence. Let $\mathbf{w} = \boldsymbol{\Pi}_K^o \mathbf{I}_K^{\text{div}} \mathbf{v}$. Subtracting the mean of $\mathbf{v}$ in (4.3), and using polynomial inverse estimates, gives

$$\|\nabla \mathbf{w}\|_K \lesssim |\mathbf{v}|_{1,K}, \quad \|\mathbf{v} - \mathbf{w}\|_K \lesssim h_K |\mathbf{v}|_{1,K}.$$

The normal interpolant trace is the face L^2 projection of $\mathbf{v} \cdot \mathbf{n}$. Thus both parts of the projected boundary jump are bounded by $\|\mathbf{v} - \mathbf{w}\|_{\partial K}$. The trace inequality now gives $\|\mathbf{v}_h\|_{1,h,K} \lesssim |\mathbf{v}|_{1,K}$. Sum over K to conclude. $\square$

266 Since b has a minus sign, use the negative of this lifting to obtain

$$267 \quad (4.5) \quad \|q_h\| \lesssim \sup_{0 \neq \mathbf{v}_h \in \mathbf{V}_h} \frac{|b(\mathbf{v}_h, q_h)|}{\|\mathbf{v}_h\|_{1,h}} \quad (q_h \in S_h).$$

268 Together with Lemmas 4.2 and 4.3, this proves that (4.1) has a unique solution [4].
 269 Also, $\operatorname{div} \mathbf{u}_h^{\operatorname{div}} \in S_h$; testing the second equation with this divergence shows that it
 270 vanishes on every element.

271 **4.2. Error equations.** Assume $(\mathbf{u}, p) \in (\mathbf{H}^{k+1}(\Omega) \cap \mathbf{H}_0^1(\Omega)) \times (H^k(\Omega) \cap L_0^2(\Omega))$
 272 solves (2.1). Set

$$273 \quad \mathbf{u}_I^{\operatorname{div}}|_K = \mathbf{I}_K^{\operatorname{div}} \mathbf{u}, \quad \mathbf{u}_I^t|_F = \Pi_{\partial K}^t \mathbf{u}, \quad p_I = Q_h p.$$

274 For later use, (4.4), polynomial approximation, and inverse estimates imply

$$275 \quad (4.6) \quad \begin{aligned} & \|\mathbf{u} - \Pi_K^o \mathbf{I}_K^{\operatorname{div}} \mathbf{u}\|_K + h_K \|\mathbf{u} - \Pi_K^o \mathbf{I}_K^{\operatorname{div}} \mathbf{u}\|_{1,K} \\ & + h_K^{1/2} \|\mathbf{u} - \Pi_K^o \mathbf{I}_K^{\operatorname{div}} \mathbf{u}\|_{\partial K} \lesssim h_K^{k+1} |\mathbf{u}|_{k+1,K}. \end{aligned}$$

276 To see this, subtract a degree- k polynomial approximant and bound the remaining
 277 polynomial by its L^2 norm using inverse and trace estimates. The same argument
 278 applied to the strain gives

$$279 \quad (4.7) \quad \begin{aligned} & \|\varepsilon(\mathbf{u}) - \varepsilon(\Pi_K^o \mathbf{I}_K^{\operatorname{div}} \mathbf{u})\|_K \\ & + h_K^{1/2} \|\varepsilon(\mathbf{u}) - \varepsilon(\Pi_K^o \mathbf{I}_K^{\operatorname{div}} \mathbf{u})\|_{\partial K} \lesssim h_K^k |\mathbf{u}|_{k+1,K}. \end{aligned}$$

280 Since the normal interpolant trace and the tangential interpolant are face projections,
 281 (4.6) also gives

$$282 \quad (4.8) \quad h_K^{-1/2} \|\Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I)\|_{\partial K} \lesssim h_K^k |\mathbf{u}|_{k+1,K}.$$

283

284 Set $e_h = \mathbf{u}_h - \mathbf{u}_I$ and $\xi_h = p_h - p_I$. The error equation is

$$285 \quad (4.9) \quad \begin{aligned} & a_h(e_h, \mathbf{v}_h) + b(\mathbf{v}_h, \xi_h) = \mathcal{R}(\mathbf{v}_h) := \sum_{K \in \mathcal{K}_h} \left[\right. \\ & \quad (\varepsilon(\mathbf{u}) - \varepsilon(\Pi_K^o \mathbf{u}_I^{\operatorname{div}}), \varepsilon(\Pi_K^o \mathbf{v}_h^{\operatorname{div}}))_K \\ & \quad + \langle \mathcal{J}_{\partial K}(\mathbf{v}_h), [\varepsilon(\mathbf{u}) - \varepsilon(\Pi_K^o \mathbf{u}_I^{\operatorname{div}})] \mathbf{n} \rangle_{\partial K} \\ & \quad + \langle Q_h p - p, (\mathbf{v}_h^{\operatorname{div}} - \Pi_K^o \mathbf{v}_h^{\operatorname{div}}) \cdot \mathbf{n} \rangle_{\partial K} \\ & \quad \left. - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \varepsilon_K^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} \right] \quad \forall \mathbf{v}_h \in \mathbf{V}_h. \end{aligned}$$

286 To derive it, test (2.1) with $\Pi_K^o \mathbf{v}_h^{\operatorname{div}}$, integrate by parts, and sum over K . The exact
 287 traction paired with $(\mathbf{v}_h^{\operatorname{div}} \cdot \mathbf{n}) \mathbf{n} + \mathbf{v}_h^t$ cancels on interior faces and vanishes on the
 288 boundary. For the pressure terms, the volume projection preserves moments against
 289 $\nabla(Q_h p)$, and

$$290 \quad (\operatorname{div} \Pi_K^o \mathbf{v}_h^{\operatorname{div}}, p - Q_h p)_K = 0.$$

291 For the strain terms, use (3.7) twice and recall that $\varepsilon(\Pi_K^o \mathbf{u}_I^{\operatorname{div}}) \in \mathbb{P}_{k-1}(K; \mathbb{S}) \subset$
 292 $\Sigma_h^{-1}(K)$. Its normal trace belongs to $V_k(F)$, so it pairs equally with the projected
 293 and full jumps. By (4.2), $\operatorname{div} \mathbf{u}_I^{\operatorname{div}} = 0$. Hence

$$294 \quad (4.10) \quad b(e_h, q_h) = 0 \quad \forall q_h \in S_h.$$

We estimate the four terms in (4.9) in order. The volume strain term is bounded by (4.7) and the norm of $\mathbf{v}_h$. The boundary strain term uses the same approximation estimate and the full-jump bound (3.8). For the pressure term, the local projection estimate is

$$h_K^{1/2} \|p - Q_h p\|_{\partial K} \lesssim h_K^k |p|_{k,K};$$

its other factor is the normal part of the jump, which the discrete norm controls directly. Finally, the polynomial trace inequality and the weak-strain upper bound give

$$\begin{aligned} & |\langle \mathbf{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \varepsilon_K^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} | \\ & \lesssim h_K^{-1/2} \|\mathbf{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I)\|_{\partial K} \|\mathbf{v}_h\|_{1,h,K}. \end{aligned}$$

Use (4.8), then sum the estimates over K . The Cauchy–Schwarz inequality yields

$$(4.11) \quad |\mathcal{R}(\mathbf{v}_h)| \lesssim h^k (\|\mathbf{u}\|_{k+1} + \|p\|_k) \|\mathbf{v}_h\|_{1,h}.$$

4.3. Error estimates. Taking $\mathbf{v}_h = e_h$ in (4.9) and $q_h = \xi_h$ in (4.10), coercivity gives

$$\|e_h\|_{1,h}^2 \lesssim a_h(e_h, e_h) = \mathcal{R}(e_h).$$

Therefore, (4.11) implies

$$\|e_h\|_{1,h} \lesssim h^k (\|\mathbf{u}\|_{k+1} + \|p\|_k).$$

For the pressure, (4.5) and (4.9) give

$$\begin{aligned} \|\xi_h\| & \lesssim \sup_{0 \neq \mathbf{v}_h \in \mathbf{V}_h} \frac{|\mathcal{R}(\mathbf{v}_h) - a_h(e_h, \mathbf{v}_h)|}{\|\mathbf{v}_h\|_{1,h}} \\ & \lesssim \|e_h\|_{1,h} + h^k (\|\mathbf{u}\|_{k+1} + \|p\|_k). \end{aligned}$$

Thus $\|\xi_h\|$ has the same bound.

In the velocity error norm, identify the smooth velocity with the pair $\{\mathbf{u}, \mathbf{\Pi}_{\partial K}^t \mathbf{u}\}$, and use the same norm formula with the L^2 volume projector. Polynomial approximation and (4.4) give $\|\mathbf{u} - \mathbf{u}_I\|_{1,h} \lesssim h^k \|\mathbf{u}\|_{k+1}$. The triangle inequality and $\|p - Q_h p\| \lesssim h^k \|p\|_k$ then yield

$$(4.12) \quad \|\mathbf{u} - \mathbf{u}_h\|_{1,h} + \|p - p_h\| \lesssim h^k (\|\mathbf{u}\|_{k+1} + \|p\|_k).$$

5. Numerical results. For $k = 1$, the velocity uses $\mathbf{V}_h$ and the pressure is piecewise constant on the primal mesh $\mathcal{K}_h$. The coefficient of the symmetric strain in (2.1) is one, which corresponds to $2\nu = 1$ in the usual viscosity convention. The weak strain (3.7) has the equivalent form

$$(\varepsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K = -(\mathbf{v}^{\text{div}}, \text{div}_{w,K} \boldsymbol{\sigma})_K + \langle \mathbf{v}^{\text{div}} \cdot \mathbf{n}, (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{n} \rangle_{\partial K} + \langle \mathbf{v}^t, \Pi_F(\boldsymbol{\sigma} \mathbf{n}) \rangle_{\partial K},$$

where $\boldsymbol{\sigma} \in \Sigma_h^{-1}(K)$ with $k = 1$, and $\text{div}_{w,K} \boldsymbol{\sigma} \in \mathbb{P}_1(K; \mathbb{R}^d)$ satisfies

$$(\text{div}_{w,K} \boldsymbol{\sigma}, \mathbf{q})_K = -(\boldsymbol{\sigma}, \nabla \mathbf{q})_K + \langle \boldsymbol{\sigma} \mathbf{n}, \mathbf{q} \rangle_{\partial K} \quad \forall \mathbf{q} \in \mathbb{P}_1(K; \mathbb{R}^d).$$

This definition includes the internal stress jumps through integration by parts on the cones. Its equivalence follows from the volume projection and the symmetry of σ .

The reported `sincosdata` test is posed on the unit square. Tables 1–3 give the supplied results. The energy error is $\|\mathbf{u}_I - \mathbf{u}_h\|_{A_h} := a_h(\mathbf{u}_I - \mathbf{u}_h, \mathbf{u}_I - \mathbf{u}_h)^{1/2}$. The energy and total pressure errors approach first-order convergence on these meshes. Faster convergence in some other columns is an empirical observation; it is not proved by (4.12).

Remark 5.1 (Numerical data to confirm). The supplied files do not specify the exact velocity, pressure, force, boundary data, viscosity used in the computations, or the definitions of $\|\cdot\|_{0,h}$ and the reported maximum norms. The implemented strain space must also be checked against (3.1); the original numerical description used a different index. The table values are retained pending these checks. The rectangular meshes do not satisfy Assumption 3.1: their face-tangential tensors span only diagonal symmetric matrices. Table 3 therefore reports a test outside the proved mesh class. The three mesh figure files were not supplied.

TABLE 1
Results on polygonal meshes obtained from a triangle-mesh dual.

DoFs	h	$\ \mathbf{u}_I - \mathbf{u}_h\ _{A_h}$	$\ \mathbf{u}_I - \mathbf{u}_h\ _{0,h}$	$\ \mathbf{u}_I - \mathbf{u}_h\ _{\infty}$	$\ p_I - p_h\ $	$\ p - p_h\ $	$\ p_I - p_h\ _{\max}$
314	1.43e-01	8.767e-01	2.588e-02	7.996e-02	2.071e-01	2.454e-01	5.002e-01
978	7.78e-02	4.012e-01	9.135e-03	2.645e-02	6.612e-02	1.007e-01	1.870e-01
3362	4.12e-02	1.884e-01	2.712e-03	7.111e-03	1.936e-02	4.463e-02	7.481e-02
12354	2.13e-02	9.119e-02	7.279e-04	1.817e-03	5.869e-03	2.145e-02	3.068e-02
47234	1.08e-02	4.490e-02	1.875e-04	4.573e-04	1.874e-03	1.061e-02	1.335e-02

TABLE 2
Results on triangular meshes.

DoFs	h	$\ \mathbf{u}_I - \mathbf{u}_h\ _{A_h}$	$\ \mathbf{u}_I - \mathbf{u}_h\ _{0,h}$	$\ \mathbf{u}_I - \mathbf{u}_h\ _{\infty}$	$\ p_I - p_h\ $	$\ p - p_h\ $	$\ p_I - p_h\ _{\max}$
232	2.50e-01	4.289e-01	1.463e-02	3.022e-02	5.357e-01	5.509e-01	8.008e-01
880	1.25e-01	1.949e-01	2.959e-03	6.035e-03	2.867e-01	2.940e-01	4.262e-01
3424	6.25e-02	9.410e-02	6.990e-04	1.479e-03	1.450e-01	1.486e-01	2.115e-01
13504	3.12e-02	4.659e-02	1.727e-04	4.138e-04	7.265e-02	7.446e-02	1.045e-01
53632	1.56e-02	2.323e-02	4.308e-05	1.098e-04	3.634e-02	3.725e-02	5.187e-02

TABLE 3
Results on rectangular meshes, outside Assumption 3.1.

DoFs	h	$\ \mathbf{u}_I - \mathbf{u}_h\ _{A_h}$	$\ \mathbf{u}_I - \mathbf{u}_h\ _{0,h}$	$\ \mathbf{u}_I - \mathbf{u}_h\ _{\infty}$	$\ p_I - p_h\ $	$\ p - p_h\ $	$\ p_I - p_h\ _{\max}$
152	2.50e-01	6.193e-01	4.377e-02	7.423e-02	1.557e-02	1.574e-01	2.227e-02
560	1.25e-01	2.438e-01	1.277e-02	1.582e-02	9.016e-03	8.021e-02	1.778e-02
2144	6.25e-02	1.083e-01	3.348e-03	4.464e-03	2.688e-03	4.011e-02	6.658e-03
8384	3.12e-02	5.211e-02	8.477e-04	1.148e-03	6.957e-04	2.004e-02	1.943e-03
33152	1.56e-02	2.579e-02	2.126e-04	2.889e-04	1.752e-04	1.002e-02	5.205e-04

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